

Practice — Pressure in Liquids and Air

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 • MCQ**[1]**

The pressure exerted by a liquid depends on the —

- (a) width of the container (b) height of the liquid column (c) colour of the liquid
(d) shape of the lid

Q2 • MCQ**[1]**

Liquids exert pressure —

- (a) only at the bottom (b) only on the sides (c) in all directions (d) only upward

Q3 • MCQ**[1]**

A rubber sucker sticks to a wall because —

- (a) it is glued (b) outside air pressure is higher than inside (c) the wall is magnetic
(d) of gravity

Q4 • MCQ**[1]**

Why is the base of a dam built broader than the top?

- (a) to look nice (b) to withstand the large water pressure near the bottom (c) to store more water
(d) to reduce its height

Q5 • TRUE / FALSE**[1]**

We are crushed by atmospheric pressure.

- (a) True (b) False

Q6 · FILL IN THE BLANK

[1]

The envelope of air surrounding the Earth is called the _____.

Q7 · FILL IN THE BLANK

[1]

The practical unit of air pressure equal to 100 Pa is the _____ (or hectopascal).

Short answer (2 marks each)

Q8

[2]

State two facts about the pressure exerted by a liquid.

Q9

[2]

What is atmospheric pressure? Why are we not crushed by it?

Q10

[2]

Explain why water spurts out from holes made on the sides of a filled bottle.

Long / application (3 marks each)

Q11

[3]

Two pipes of different diameters are filled with water to the same height; balloons fixed at their lower ends bulge equally. Explain why, and what happens if more water is added to one pipe.

Q12

[3]

Describe an activity showing that air exerts pressure, and explain the observation.

[Check yourself](#) → Answer key